

Your
Team
Name

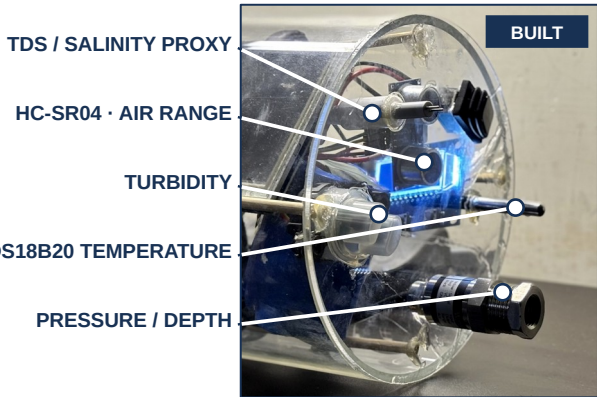
AquaSDR • TECHNICAL APPROACH

Adaptive Software-Defined Sonar Transmitter for AUVs



SENSE

Real environment • real sensors



ADC • 1-Wire • I²C

Also on board BMP390 • IMU + magnetometer • INA219 ×2
(mounted, not commissioned)

ONE POLICY • TWO INPUTS



DECIDE • SYNTHESISE • STREAM

STM32F446RE embedded control

THE NEXT PING IS COMPUTED ON THE MCU

01 ACQUIRE
ADC • 1-Wire • I²C
Reject stale / invalid inputs

02 DECIDE
Smoothing + hysteresis
CLEAR / TRANSITION / MURKY

03 SYNTHESISE
LFM • Geometric • Barker-13
Hann • Hamming • Blackman

04 STREAM
TIM6 + DMA → 12-bit DAC1 / PA4
1 MS/s • 0 underruns • CPU free between transfers

06 SELF-VALIDATE
ADC + DMA capture of DAC output
FFT • spectrogram • matched filter

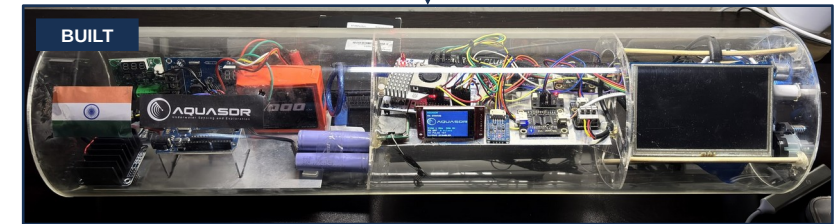
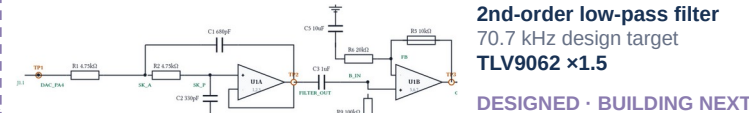
ADAPTED NEXT PING

Adaptive waveform parameters

	CLEAR	TRANSITION	MURKY
CENTRE	42 kHz	40 kHz	38 kHz
BANDWIDTH	4 kHz	2 kHz	1 kHz
PULSE	2 ms	4 ms	8 ms
AMPLITUDE	35 %	50 %	62 %

CLEAR → shorter / sharper ping
MURKY → lower band / longer pulse / higher output demand

05 CONDITION • ANALOG FRONT END



AQUASDR PAYLOAD • PHYSICAL PROTOTYPE acrylic hull

Scope validation → PENDING • AUV hull-slot pod → NEXT BUILD

TECHNOLOGIES TO BE USED	Embedded	STM32F446RE • C/C++ • ADC • TIM6 • DMA • DAC	Signal processing	LFM • Geometric • Barker-13 • FFT • Windowing
	Analog	Sallen-Key LPF • TLV9062	Software	React • Node.js • WebSocket • Python • OpenSCAD



— built / running - - - - - designed / building next